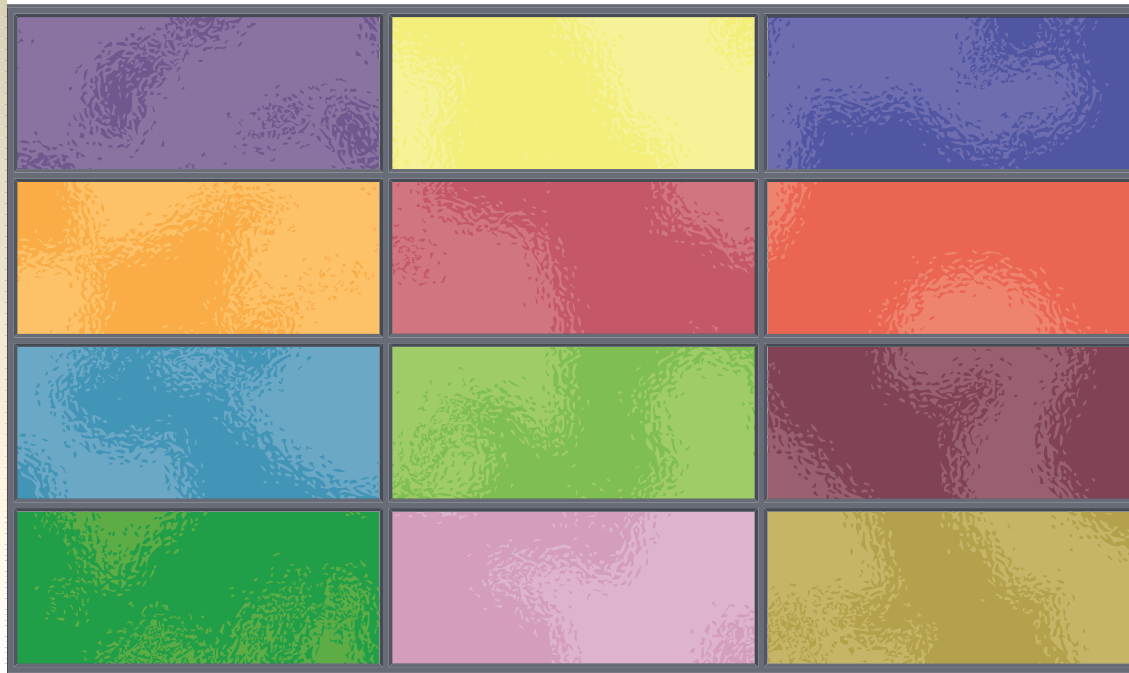


3.7 Multiplying Polynomials

LESSON FOCUS

Extend the strategies for multiplying binomials to multiplying polynomials.



Make Connections

The art work above was a design for a stained glass panel. How could you use the length and width of each small rectangle to determine the area of the large rectangle without first finding its overall length and width?

Construct Understanding

THINK ABOUT IT

Work with a partner.

- Draw a rectangle diagram to determine the product $(a + b + 2)(c + d + 3)$.
- Draw a rectangle diagram to determine the product $(a - b + 2)(c + d - 3)$.
- Use a different strategy to check the products.

Would the product have been different if you reversed the order of the factors? Explain.

The distributive property can be used to perform any polynomial multiplication. Each term of one polynomial must be multiplied by each term of the other polynomial.

Example 1**Using the Distributive Property to Multiply Two Polynomials**

Expand and simplify.

a) $(2h + 5)(h^2 + 3h - 4)$ b) $(-3f^2 + 3f - 2)(4f^2 - f - 6)$

SOLUTION

- a) Use the distributive property. Multiply each term in the trinomial by each term in the binomial.

Write the terms in a list.

$$\begin{aligned} & (2h + 5)(h^2 + 3h - 4) \\ &= (2h)(h^2 + 3h - 4) + 5(h^2 + 3h - 4) \\ &= (2h)(h^2) + (2h)(3h) + 2h(-4) + 5(h^2) + 5(3h) + 5(-4) \\ &= 2h^3 + 6h^2 - 8h + 5h^2 + 15h - 20 \\ &= 2h^3 + 6h^2 + 5h^2 - 8h + 15h - 20 \quad \text{Combine like terms.} \\ &= 2h^3 + 11h^2 + 7h - 20 \end{aligned}$$

- b) Use the distributive property. Multiply each term in the 1st trinomial by each term in the 2nd trinomial.
Align like terms.

$$\begin{array}{r} (-3f^2 + 3f - 2)(4f^2 - f - 6): \\ -3f^2(4f^2 - f - 6): \quad -12f^4 + 3f^3 + 18f^2 \\ 3f(4f^2 - f - 6): \quad \quad \quad 12f^3 - 3f^2 - 18f \\ -2(4f^2 - f - 6): \quad \quad \quad \quad -8f^2 + 2f + 12 \\ \hline \text{Add:} \quad \quad \quad -12f^4 + 15f^3 + 7f^2 - 16f + 12 \end{array}$$

CHECK YOUR UNDERSTANDING

1. Expand and simplify.

a) $(3k + 4)(k^2 - 2k - 7)$

b) $(-2t^2 + 4t - 3)(5t^2 - 2t + 1)$

[Answers: a) $3k^3 - 2k^2 - 29k - 28$

b) $-10t^4 + 24t^3 - 25t^2 + 10t - 3$]

Both solutions to parts a and b use the distributive property. Which strategy for recording the products of terms do you prefer, and why?

One way to check that a product is likely correct is to substitute a number for the variable in both the trinomial product and its simplification. If both expressions are equal, the product is likely correct.

In *Example 1a*, substitute $h = 1$.

$$(2h + 5)(h^2 + 3h - 4) = 2h^3 + 11h^2 + 7h - 20$$

$$\begin{aligned} \text{Left side: } (2h + 5)(h^2 + 3h - 4) &= [2(1) + 5][(1)^2 + 3(1) - 4] \\ &= (7)(0) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Right side: } 2h^3 + 11h^2 + 7h - 20 &= 2(1)^3 + 11(1)^2 + 7(1) - 20 \\ &= 2 + 11 + 7 - 20 \\ &= 0 \end{aligned}$$

Since the left side equals the right side, the product is likely correct.

In *Example 1b*, substitute $f = 1$.

$$(-3f^2 + 3f - 2)(4f^2 - f - 6) = -12f^4 + 15f^3 + 7f^2 - 16f + 12$$

$$\text{Left side: } (-3f^2 + 3f - 2)(4f^2 - f - 6)$$

$$= [(-3)(1)^2 + 3(1) - 2] [(4(1)^2 - (1) - 6)]$$

$$= (-2)(-3)$$

$$= 6$$

$$\text{Right side: } -12f^4 + 15f^3 + 7f^2 - 16f + 12$$

$$= -12(1)^4 + 15(1)^3 + 7(1)^2 - 16(1) + 12$$

$$= -12 + 15 + 7 - 16 + 12$$

$$= 6$$

Since the left side equals the right side, the product is likely correct.

Example 2

Multiplying Polynomials in More than One Variable

Expand and simplify.

a) $(2r + 5t)^2$ b) $(3x - 2y)(4x - 3y + 5)$

SOLUTION

a) $(2r + 5t)^2 = (2r + 5t)(2r + 5t)$

$$= 2r(2r + 5t) + 5t(2r + 5t)$$

$$= 2r(2r) + 2r(5t) + 5t(2r) + 5t(5t)$$

$$= 4r^2 + 10rt + 10rt + 25t^2 \quad \text{Combine like terms.}$$

$$= 4r^2 + 20rt + 25t^2$$

b) $(3x - 2y)(4x - 3y + 5)$

$$= 3x(4x - 3y - 5) - 2y(4x - 3y + 5)$$

$$= 3x(4x) + 3x(-3y) + 3x(-5) - 2y(4x) - 2y(-3y) - 2y(5)$$

$$= 12x^2 - 9xy - 15x - 8xy + 6y^2 - 10y \quad \text{Collect like terms.}$$

$$= 12x^2 - 9xy - 8xy - 15x + 6y^2 - 10y \quad \text{Combine like terms.}$$

$$= 12x^2 - 17xy - 15x + 6y^2 - 10y$$

CHECK YOUR UNDERSTANDING

2. Expand and simplify.

a) $(4k - 3m)^2$

b) $(2v - 5w)(3v + 2w - 7)$

[Answers: a) $16k^2 - 24km + 9m^2$

b) $6v^2 - 11vw - 14v - 10w^2 + 35w$]

Look at the factors and product in part a. How could you write the product without using the distributive property?

Example 3**Simplifying Sums and Differences of Polynomial Products**

Expand and simplify.

a) $(2c - 3)(c + 5) + 3(c - 3)(-3c + 1)$

b) $(3x + y - 1)(2x - 4) - (3x + 2y)^2$

SOLUTION

Use the order of operations. Multiply before adding and subtracting. Then combine like terms.

a) $(2c - 3)(c + 5) + 3(c - 3)(-3c + 1)$

$$= 2c(c + 5) - 3(c + 5) + 3[c(-3c + 1) - 3(-3c + 1)]$$

$$= 2c^2 + 10c - 3c - 15 + 3[-3c^2 + c + 9c - 3]$$

$$= 2c^2 + 7c - 15 + 3[-3c^2 + 10c - 3]$$

$$= 2c^2 + 7c - 15 - 9c^2 + 30c - 9$$

$$= -7c^2 + 37c - 24$$

b) $(3x + y - 1)(2x - 4) - (3x + 2y)^2$

$$= 3x(2x - 4) + y(2x - 4) - 1(2x - 4) - (3x + 2y)(3x + 2y)$$

$$= 6x^2 - 12x + 2xy - 4y - 2x + 4 - [3x(3x + 2y) + 2y(3x + 2y)]$$

$$= 6x^2 - 14x + 2xy - 4y + 4 - [9x^2 + 6xy + 6xy + 4y^2]$$

$$= 6x^2 - 14x + 2xy - 4y + 4 - [9x^2 + 12xy + 4y^2]$$

$$= 6x^2 - 14x + 2xy - 4y + 4 - 9x^2 - 12xy - 4y^2$$

$$= -3x^2 - 14x - 10xy - 4y + 4 - 4y^2$$

CHECK YOUR UNDERSTANDING

3. Expand and simplify.

a) $(4m + 1)(3m - 2) + 2(2m - 1)(-3m + 4)$

b) $(6h + k - 2)(2h - 3) - (4h - 3k)^2$

[Answers: a) $17m - 10$

b) $-4h^2 - 22h + 26hk - 3k + 6 - 9k^2$]

Discuss the Ideas

1. How is the process for multiplying polynomials with more than 2 terms like multiplying binomials? How is it different?
2. What strategies can you use to check that you have correctly multiplied two polynomials?
3. When you substitute a number for a variable to check that a polynomial product is correct, would 0 be a suitable number to substitute? Would 9 be a suitable number? Explain.

Exercises

A

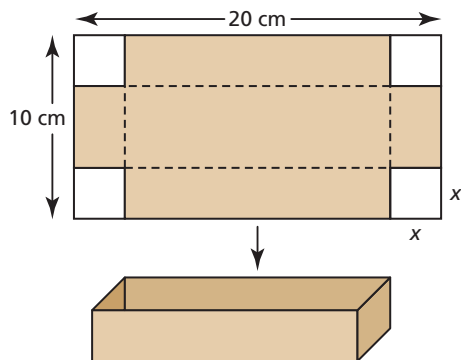
4. Expand and simplify.
- $(g + 1)(g^2 + 2g + 3)$
 - $(2 + t + t^2)(1 + 3t + t^2)$
 - $(2w + 3)(w^2 + 4w + 7)$
 - $(4 + 3n + n^2)(3 + 5n + n^2)$
5. Expand and simplify.
- $(2z + y)(3z + y)$
 - $(4f - 3g)(3f - 4g + 1)$
 - $(2a + 3b)(4a + 5b)$
 - $(3a - 4b + 1)(4a - 5b)$
 - $(2r + s)^2$
 - $(3t - 2u)^2$

B

6. a) Expand and simplify.
- $(2x + y)(2x + y)$
 - $(5r + 2s)(5r + 2s)$
 - $(6c + 5d)(6c + 5d)$
 - $(5v + 7w)(5v + 7w)$
 - $(2x - y)(2x - y)$
 - $(5r - 2s)(5r - 2s)$
 - $(6c - 5d)(6c - 5d)$
 - $(5v - 7w)(5v - 7w)$
- b) What patterns do you see in the factors and products in part a? Use these patterns to expand and simplify each product without using the distributive property.
- $(p + 3q)(p + 3q)$
 - $(2s - 7t)(2s - 7t)$
 - $(5g + 4h)(5g + 4h)$
 - $(10h - 7k)(10h - 7k)$
7. a) Expand and simplify.
- $(x + 2y)(x - 2y)$
 - $(3r - 4s)(3r + 4s)$
 - $(5c + 3d)(5c - 3d)$
 - $(2v - 7w)(2v + 7w)$
- b) What patterns do you see in the factors and products in part a? Use these patterns to expand and simplify each product without using the distributive property.
- $(11g + 5h)(11g - 5h)$
 - $(25m - 7n)(25m + 7n)$

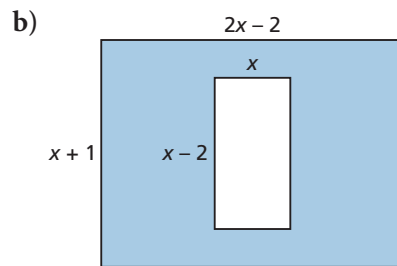
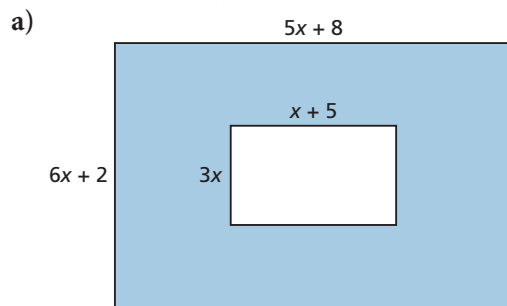
8. Expand and simplify.
- $(3y - 2)(y^2 + y - 8)$
 - $(4r + 1)(r^2 - 2r - 3)$
 - $(b^2 + 9b - 2)(2b - 1)$
 - $(x^2 + 6x + 1)(3x - 7)$
9. Expand and simplify.
- $(x + y)(x + y + 3)$
 - $(x + 2)(x + y + 1)$
 - $(a + b)(a + b + c)$
 - $(3 + t)(2 + t + s)$
10. Expand and simplify.
- $(x + 2y)(x - 2y - 1)$
 - $(2c - 3d)(c + d + 1)$
 - $(a - 5b)(a + 2b - 4)$
 - $(p - 2q)(p + 4q - r)$
11. Find and correct the errors in this solution.
- $$\begin{aligned} & (2r - 3s)(r - 5s + 6) \\ &= 2r(r - 5s + 6) - 3s(r - 5s + 6) \\ &= 2r^2 - 5rs + 12r - 3rs - 15s^2 - 18s \\ &= 2r^2 - 8rs + 12r - 33s^2 \end{aligned}$$
12. The area of the base of a right rectangular prism is $x^2 + 3x + 2$. The height of the prism is $x + 7$. Write, then simplify an expression for the volume of the prism.
13. Expand and simplify. Substitute a number for the variable to check each product.
- $(r^2 + 3r + 2)(4r^2 + r + 1)$
 - $(2d^2 + 2d + 1)(d^2 + 6d + 3)$
 - $(4c^2 - 2c - 3)(-c^2 + 6c + 2)$
 - $(-4n^2 - n + 3)(-2n^2 + 5n - 1)$
14. Find and correct the errors in this solution.
- $$\begin{aligned} & (3g^2 + 4g - 2)(-g^2 - g + 4) \\ &= -3g^4 - 3g^3 + 12g^2 - 4g^3 + 4g^2 + 8g \\ & \quad + 2g^2 + 2g + 8 \\ &= -3g^4 + 5g^3 + 6g^2 + 10g + 8 \end{aligned}$$
15. Expand and simplify.
- $(3s + 5)(2s + 2) + (3s + 7)(s + 6)$
 - $(2x + 3)(5x + 4) + (x - 4)(3x - 7)$
 - $(3m + 4)(m - 4n) + (5m - 2)(3m - 6n)$
 - $(4y - 5)(3y + 2) - (3y + 2)(4y - 5)$
 - $(3x - 2)^2 - (2x + 6)(3x - 1)$
 - $(2a + 1)(4a - 3) - (a - 2)^2$

16. A box with no top is made from a piece of cardboard 20 cm by 10 cm. Equal squares are cut from each corner and the sides are folded up.



Let x centimetres represent the side length of each square cut out. Write a polynomial to represent each measurement. Simplify each polynomial.

- the length of the box
 - the width of the box
 - the area of the base of the box
 - the volume of the box
17. Each shape is a rectangle. Write a polynomial to represent the area of each shaded region. Simplify each polynomial.



C

18. Expand and simplify.
- $(x - 2)^3$
 - $(2y + 5)^3$
 - $(4a - 3b)^3$
 - $(c + d)^3$
19. Expand and simplify.
- $2a(2a - 1)(3a + 2)$
 - $-3r(r - 1)(2r + 1)$
 - $5x^2(2x - 1)(4x - 3)$
 - $-xy(2x + 5)(4x - 5)$
 - $2b(2b - c)(b + c)$
 - $y^2(y^2 + 1)(y^2 - 1)$
20. A cube has edge length $2x + 3$.
- Write then simplify an expression for the volume of the cube.
 - Write then simplify an expression for the surface area of the cube.
21. Expand and simplify.
- $(3x + 4)(x - 5)(2x + 8)$
 - $(b - 7)(b + 8)(3b - 4)$
 - $(2x - 5)(3x + 4)^2$
 - $(5a - 3)^2(2a - 7)$
 - $(2k - 3)(2k + 3)^2$
22. Expand and simplify.
- $(x + y + 1)^3$
 - $(x - y - 1)^3$
 - $(x + y + z)^3$
 - $(x - y - z)^3$

Reflect

What strategies do you know for multiplying two binomials? How can you use or adapt those strategies to multiply two trinomials? Include examples in your explanation.